

Diploma in Electronics / Electronics & Communication  
Engineering • Semester 3

# Electric Circuits & Networks

Full-width desktop lesson sample with textbook-style explanation, diagrams, formula cards, Malayalam support, module-wise questions, and a matching Master Answer Key. Only one selected section loads on screen.

Course Code: 3041

Category: Program Core

Credits: 3

L:T:P = 2:1:0

Theory Subject

## Lesson structure check

4

Official modules

45

Periods / semester

A4

Print-ready layout

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## Course Overview

**Subject title:** Electric Circuits & Networks

**Semester:** 3 | **Course category:** Program Core | **Subject type:** Engineering Theory subject

This subject connects basic electrical laws, AC circuit behavior, network analysis, transformers, DC machines and AC machines. A student who studies this subject properly can read simple circuit diagrams, calculate circuit quantities, apply network theorems, and explain the working of common electrical machines used in electronics and industry.

**Malayalam support:** ് subject-ൿ electric circuit-ൿൿൿൿ behavior, AC signal, RLC circuit, network theorem, transformer, DC machine, AC machine ൿൿൿൿൿ ൿൿൿൿൿൿൿൿൿൿ. Technical terms English-ൿ ൿൿൿൿൿ ൿൿൿൿൿൿൿ exam answer ൿൿൿൿൿ ൿൿൿൿൿൿൿൿൿൿ.

Source basis: Official SITTTTR Revision 2021 syllabus content for course code 3041.

## Course objective

To develop the skill to diagnose and rectify electric circuit networks and understand the operations of electrical machines.

**Simple meaning:** Circuit fault ൿൿൿൿൿൿ ൿൿൿൿൿൿൿൿൿൿൿ, circuit values ൿൿൿൿൿൿ calculate ൿൿൿൿൿൿ, motor/transformer working ൿൿൿൿൿൿ explain ൿൿൿൿൿൿ — ൿൿൿൿൿ main aim.

## Prerequisite knowledge

- Basic Engineering Mathematics principles from Mathematics I & II.
- Magnetism concepts from Applied Physics I & II.
- Ohm's law, units, algebra, trigonometry and simple vector/phaser idea.

## Course outcomes

| CO      | Outcome                                                                                      | Durati<br>on | Level    |
|---------|----------------------------------------------------------------------------------------------|--------------|----------|
| CO<br>1 | Solve a given AC circuit to find various parameters using AC signals and component behavior. | 9 hours      | Applying |

| CO   | Outcome                                                                               | Duration | Level         |
|------|---------------------------------------------------------------------------------------|----------|---------------|
| CO 2 | Apply network theorems for simplifying circuits and illustrate transformer operation. | 14 hours | Applying      |
| CO 3 | Illustrate principles and operations of DC machines.                                  | 10 hours | Understanding |
| CO 4 | Illustrate principles and operations of AC machines.                                  | 10 hours | Understanding |

## Module map

### Module 1

AC signals, R-L-C behavior, phasor diagram, AC power, resonance, series and parallel RLC circuits.

### Module 2

Ohm's law, Kirchhoff's laws, mesh and node analysis, network theorems, transformer principle and losses.

### Module 3

DC generator, DC motor, back emf, starter, 3-point starter, DC motor comparison and characteristics.

### Module 4

Alternator, AC motor classification, stepper motor, servo motor, universal motor, single-phase and three-phase induction motor.

## How to study this subject

- 1 First learn definitions and units: RMS value, average value, frequency, reactance, impedance, power factor, resonance.
- 2 Draw every circuit and machine diagram in a separate notebook. Label current direction, voltage, winding, core and load.
- 3 For numerical problems, always write Given → Formula → Substitution → Calculation → Final answer with unit.
- 4 For theorem questions, write statement, circuit before simplification, circuit after simplification, calculation steps and result.
- 5 For machine questions, write principle, construction, working, equation, characteristics and applications.

## Exam preparation method

### Short answers

Definitions, symbols, units, laws, types, applications and advantages.

### Problem solving

Practice AC power, impedance, resonance, Thevenin equivalent, maximum power transfer and transformer ratio.

### Long answers

Use diagram + working + equation + application format. This gives better marks.

**Malayalam tip:** Diagram ചുരുക്കം question ചുരുക്കം diagram ചുരുക്കം ചുരുക്കം. ചുരുക്കം working principle ചുരുക്കം. Formula ചുരുക്കം ചുരുക്കം, explanation-ചുരുക്കം ചുരുക്കം ചുരുക്കം.

## Quick navigation

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- Not available yet
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Desktop layout uses 98vw up to 1760px, so it fills laptop/desktop screen instead of staying small.

## Mini Formula Bank

Use this as the first revision page before solving problems. Formula meaning and unit are important for marks.

### Frequency

$$f = 1/T$$

f in Hz, T in seconds.

### Angular frequency

$$\omega = 2\pi f$$

Used in reactance formulas.

### RMS value

$$V_{\text{rms}} = V_{\text{m}}/\sqrt{2}$$

For sinusoidal waveform.

### Average value

$$V_{\text{avg}} = 0.637V_{\text{m}}$$

Average of half cycle sine wave.

### Inductive reactance

$$X_{\text{L}} = 2\pi fL$$

Increases with frequency.

### Capacitive reactance

$$X_{\text{C}} = 1/(2\pi fC)$$

Decreases with frequency.

### Series RLC impedance

$$Z = \sqrt{R^2 + (X_{\text{L}} - X_{\text{C}})^2}$$

Unit: ohm.

### Power factor

$$\cos\phi = R/Z$$

Lagging for inductive circuit.

### Active power

$$P = VI\cos\phi$$

Unit: watt.

### Reactive power

$$Q = VI\sin\phi$$

Unit: VAR.

### Apparent power

$$S = VI$$

Unit: VA.

### Resonant frequency

$$f_r = 1/(2\pi\sqrt{LC})$$

At resonance  $X_L = X_C$ .

### Thevenin resistance

$$R_{th}$$

Equivalent resistance seen from load terminals.

### Maximum power

$$R_L = R_{th}$$

Condition for maximum power transfer.

### Transformer ratio

$$V_1/V_2 = N_1/N_2$$

Ideal transformer relation.

### Transformer EMF

$$E = 4.44 f N \Phi_m$$

RMS induced emf.

### DC generator EMF

$$E = P\Phi ZN / 60A$$

Generated emf equation.

### Back EMF

$$E_b = V - I_a R_a$$

For DC motor.

### Alternator EMF

$$E = 4.44 k_w f \Phi T$$

Per phase emf.

### Synchronous speed

$$N_s = 120f/P$$

rpm; P = number of poles.

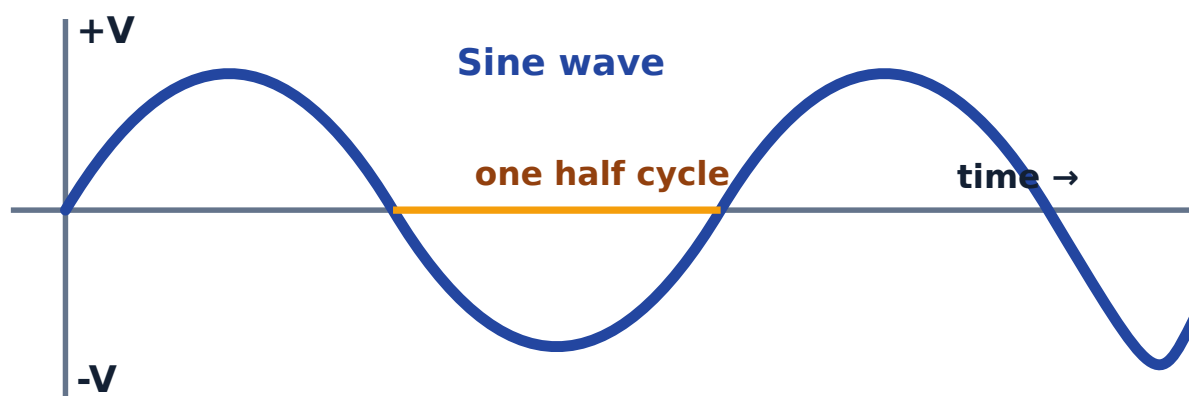
## Module 1 — AC Signals and RLC Circuits

**Learning outcome:** Solve a given AC circuit to find various parameters using the concept of AC signals and the behavior of AC through different components.

**Malayalam support:** AC circuit problem തിരിച്ചറിയുന്ന waveform, RMS, average value, reactance, impedance, phase angle, power factor തിരിച്ചറിയുന്ന clear തിരിച്ചറിയുന്ന. R, L, C തിരിച്ചറിയുന്ന current-തിരിച്ചറിയുന്ന തിരിച്ചറിയുന്ന തിരിച്ചറിയുന്ന.

### 1.1 Alternating quantities

An alternating voltage or current changes its magnitude and direction periodically. A sine wave is the most common AC waveform because generation in alternators naturally gives sinusoidal voltage.



AC sine waveform: one complete cycle repeats after time period T.

#### Important definitions

- **Cycle:** One complete set of positive and negative values.
- **Time period:** Time taken for one cycle.



- **Frequency:** Number of cycles per second.
- **Amplitude:** Maximum value of the alternating quantity.
- **Phase:** Angular position of a waveform with respect to reference.
- **Form factor:** RMS value divided by average value.

## 1.2 AC through R, L and C

| Element          | Current-voltage relation              | Opposition          | Power behavior                   |
|------------------|---------------------------------------|---------------------|----------------------------------|
| Pure Resistance  | Voltage and current are in phase.     | $R$                 | Consumes active power.           |
| Pure Inductance  | Current lags voltage by $90^\circ$ .  | $X_L = 2\pi fL$     | Stores energy in magnetic field. |
| Pure Capacitance | Current leads voltage by $90^\circ$ . | $X_C = 1/(2\pi fC)$ | Stores energy in electric field. |

**Memory trick:** Inductor- current lags voltage:  $L = \text{Lag}$ . Capacitor- current leads voltage:  $C = \text{Current leads}$ .

## 1.3 Series RLC circuit

In a series RLC circuit, the same current flows through R, L and C. The total opposition is called impedance. The inductive and capacitive reactances oppose each other, so the net reactance is  $X_L - X_C$ .



**Series RLC circuit: same current flows through all components.**

## 1.4 Solved example format

### Example: Find impedance of a series RLC circuit

**Given:**  $R = 30\ \Omega$ ,  $X_L = 50\ \Omega$ ,  $X_C = 10\ \Omega$ .

**Formula:**  $Z = \sqrt{R^2 + (X_L - X_C)^2}$

**Substitution:**  $Z = \sqrt{30^2 + (50 - 10)^2} = \sqrt{900 + 1600}$

**Calculation:**  $Z = \sqrt{2500} = 50\ \Omega$

**Final answer:** Impedance =  $50\ \Omega$ .

## Expected Questions — Module 1

**M1-Q1.** Define cycle, time period, frequency, amplitude and phase of an AC waveform.

**M1-Q2.** Compare sine, square, triangular and sawtooth waveforms.

**M1-Q3.** Explain AC through pure resistance, pure inductance and pure capacitance.

**M1-Q4.** Draw and explain the phasor diagram of a series RLC circuit.

**M1-Q5.** Define active power, reactive power, apparent power and power factor.

**M1-Q6.** A series RLC circuit has  $R = 40\ \Omega$ ,  $X_L = 80\ \Omega$  and  $X_C = 50\ \Omega$ . Find impedance and power factor.

**M1-Q7.** Explain resonance and Q factor in an RLC circuit.

**M1-Q8.** List the steps to solve a parallel AC circuit problem.

# Module 1 checklist

- AC terms
- RMS
- Reactance
- Impedance
- Power factor
- Resonance

**Common mistake:** Do not add  $X_L$  and  $X_C$  directly in series RLC. Use net reactance  $X_L - X_C$ .

# Module 2 — Network Theorems and Transformers

**Learning outcome:** Apply network theorems for simplifying electric circuits and illustrate transformer operation.

**Malayalam support:** ഏതെങ്കിലും circuit-ന്റെ equivalent circuit നൽകി solve ചെയ്യാൻ network theorem-ന്റെ main use. Transformer AC voltage step-up/step-down നൽകി വിശദീകരിക്കുക.

## 2.1 Laws and analysis methods

| Method                  | Use                                      | Exam writing point                                   |
|-------------------------|------------------------------------------|------------------------------------------------------|
| Ohm's law               | Relates voltage, current and resistance. | $V = IR$                                             |
| Kirchhoff's Current Law | Current relation at a node.              | Sum of incoming currents = sum of outgoing currents. |
| Kirchhoff's Voltage Law | Voltage relation in a closed loop.       | Algebraic sum of voltages in a loop is zero.         |
| Mesh analysis           | Solves circuits by loop currents.        | Best for planar circuits with loops.                 |
| Node analysis           | Solves circuits by node voltages.        | Best when many current sources are present.          |

## 2.2 Network theorems

### Superposition theorem

In a linear bilateral network, current or voltage in an element is the algebraic sum of effects due to each independent source acting alone.

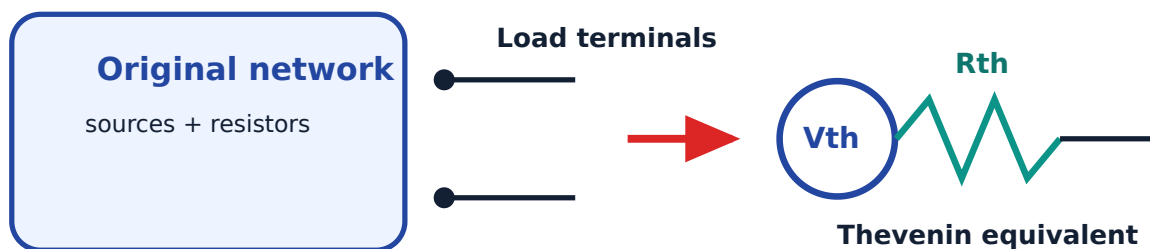
### Thevenin's theorem

Any linear two-terminal network can be replaced by a voltage source  $V_{th}$  in series with resistance  $R_{th}$ .

### Maximum power transfer

Maximum power is transferred to the load when load resistance equals Thevenin resistance.

## 2.3 Thevenin equivalent visual



Thevenin theorem replaces a complex network with  $V_{th}$  and  $R_{th}$ .

## 2.4 Transformer basics

A transformer is a static AC device that transfers electrical energy from one circuit to another by mutual induction. It can increase or decrease AC voltage without changing frequency.

### Construction

- Laminated magnetic core
- Primary winding
- Secondary winding
- Insulation and tank/cooling arrangement

### Losses

- Copper loss in windings
- Hysteresis loss in core
- Eddy current loss in core
- Leakage flux related losses

**Malayalam support:** Transformer DC- $\phi$  work  $\phi$   $\frac{d\phi}{dt}$  mutual induction  $\frac{d\phi}{dt}$  changing flux  $\phi$   $\frac{d\phi}{dt}$ . DC steady  $\phi$   $\frac{d\phi}{dt}$  changing flux  $\phi$   $\frac{d\phi}{dt}$ .

## Expected Questions — Module 2

**M2-Q1.** State Ohm's law, Kirchhoff's Current Law and Kirchhoff's Voltage Law.

**M2-Q2.** Compare mesh analysis and node analysis.

**M2-Q3.** State and explain Superposition theorem.

**M2-Q4.** State Thevenin's theorem and write the steps to find Thevenin equivalent circuit.

**M2-Q5.** State Maximum Power Transfer theorem and its condition.

**M2-Q6.** Explain the working principle and construction of a transformer.

**M2-Q7.** Derive the emf equation of a transformer.

**M2-Q8.** List transformer losses, types and applications.

## Module 2 checklist

Ohm

KCL

KVL

Mesh

Node

Thevenin

Transformer

For theorem answers: statement → circuit → steps → result → application.

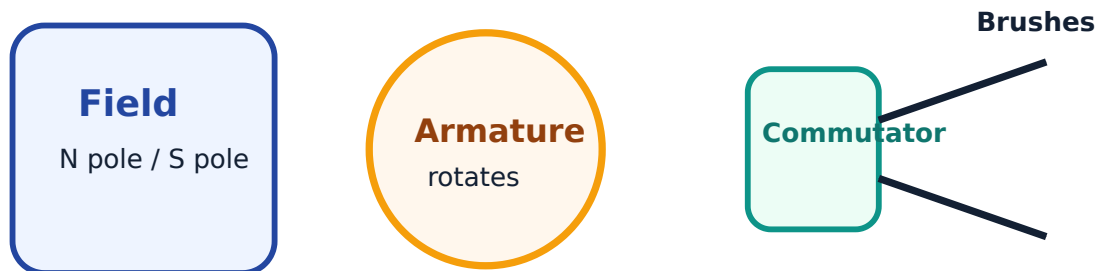
## Module 3 — DC Machines

**Learning outcome:** Illustrate the principles and operations of DC generators and DC motors.

**Malayalam support:** DC generator mechanical energy-→ DC electrical energy →→→→→. DC motor DC electrical energy-→ mechanical energy →→→→→. →→→→→ magnetic field + conductor + motion main idea →→.

### 3.1 DC generator

A DC generator works on Faraday's law of electromagnetic induction. When an armature conductor cuts magnetic flux, emf is induced in the conductor. The commutator converts the internally generated alternating emf into unidirectional output at the brushes.



**Mechanical input → DC electrical output**

**Basic blocks of a DC generator.**

## 3.2 DC motor

A DC motor works on the principle that a current-carrying conductor placed in a magnetic field experiences a mechanical force. Back emf is generated in the armature and opposes the applied voltage. At starting, back emf is zero, so starter resistance is required to limit armature current.

| Motor type               | Characteristic                                    | Common application                 |
|--------------------------|---------------------------------------------------|------------------------------------|
| <b>DC shunt motor</b>    | Nearly constant speed                             | Lathes, fans, blowers              |
| <b>DC series motor</b>   | High starting torque                              | Traction, cranes, hoists           |
| <b>DC compound motor</b> | Good starting torque with better speed regulation | Presses, elevators, heavy machines |

## 3.3 Starter in DC motor

A starter is used to limit the high starting current of a DC motor. It also provides no-voltage protection and overload protection in practical systems. A 3-point starter has line, armature and field terminals.

**Common exam mistake:** Do not write that starter increases speed. The main purpose

is to limit starting current and protect the motor.

## Expected Questions — Module 3

**M3-Q1.** Explain the working principle of a DC generator.

**M3-Q2.** Write the emf equation of a DC generator and explain each term.

**M3-Q3.** Explain armature reaction in a DC generator.

**M3-Q4.** Explain the working principle of a DC motor.

**M3-Q5.** What is back emf? Explain its significance in a DC motor.

**M3-Q6.** Why is a starter necessary in a DC motor?

**M3-Q7.** Explain the working of a 3-point starter.

**M3-Q8.** Compare DC shunt, series and compound motors with characteristics and applications.

## Module 3 checklist

Generator

EMF equation

Armature reaction

Motor

Back EMF

Starter

## Module 4 — AC Machines

**Learning outcome:** Illustrate the principles and operations of alternators and AC motors.



**Malayalam support:** Alternator AC generate ഐക്യചാലകം. AC motors AC supply ഐക്യചാലകം rotating magnetic field ഐക്യചാലകം mechanical output ഐക്യചാലകം.

## 4.1 Alternator

An alternator converts mechanical energy into AC electrical energy. The rotor produces magnetic field and the stator contains armature windings. When the rotor rotates, the magnetic flux linking the stator conductors changes and emf is induced.

### Important equations

- Frequency relation:  $f = PN / 120$
- Synchronous speed:  $N_s = 120f / P$
- EMF equation:  $E = 4.44 k_w f \Phi T$

### Open circuit characteristic

OCC shows the relation between generated emf and field current at constant speed. It is also called magnetization characteristic.

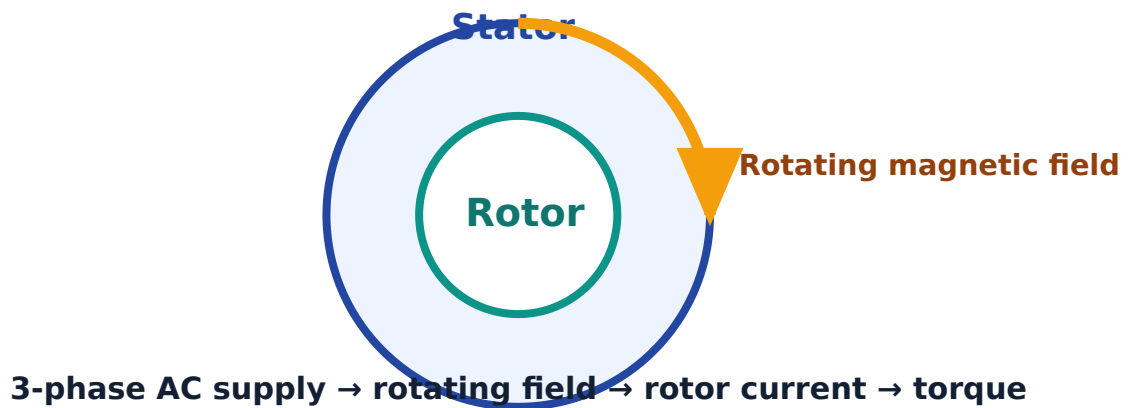
## 4.2 AC motors

| Motor                        | Working idea                                             | Application                            |
|------------------------------|----------------------------------------------------------|----------------------------------------|
| Stepper motor                | Moves in fixed angular steps.                            | Printers, CNC positioning, robotics.   |
| Servo motor                  | Feedback-controlled motor for precise position or speed. | Automation, antenna control, robotics. |
| Universal motor              | Can work on AC or DC; high speed.                        | Drills, mixers, vacuum cleaners.       |
| Single-phase induction motor | Uses single-phase supply; needs starting arrangement.    | Fans, pumps, small machines.           |

| Motor                              | Working idea                                                 | Application                                |
|------------------------------------|--------------------------------------------------------------|--------------------------------------------|
| <b>Three-phase induction motor</b> | Uses rotating magnetic field produced by three-phase supply. | Industrial drives, compressors, conveyors. |

## 4.3 Three-phase induction motor principle

When a balanced three-phase supply is given to the stator winding, a rotating magnetic field is produced. This rotating field cuts the rotor conductors and induces emf. Since the rotor circuit is closed, current flows in the rotor. Interaction between the rotating field and rotor current produces torque.



Working principle of a three-phase induction motor.

## Expected Questions — Module 4

**M4-Q1.** Explain the working principle of an alternator.

**M4-Q2.** Write the emf equation of an alternator and explain the terms.

**M4-Q3.** Define synchronous speed and derive the relation  $N_s = 120f/P$ .

**M4-Q4.** What is open circuit characteristic of an alternator?

**M4-Q5.** Classify AC motors.

**M4-Q6.** Explain the working and applications of stepper motor, servo motor and universal motor.

**M4-Q7.** Explain the working principle of a single-phase induction motor.

**M4-Q8.** Explain the working principle and applications of a three-phase induction motor.

## Module 4 checklist

Alternator

EMF

Ns

OCC

Stepper

Servo

Induction motor

## One-page Quick Revision

### Module 1 must-know

- AC terms and waveforms
- RMS and average values
- R, L and C behavior
- Impedance, power factor and resonance

### Module 2 must-know

- Ohm's law, KCL, KVL
- Mesh and node analysis
- Superposition, Thevenin, maximum power transfer
- Transformer principle, emf equation, losses and applications

## Module 3 must-know

- DC generator principle and emf equation
- Armature reaction
- DC motor principle and back emf
- Need of starter and 3-point starter

## Module 4 must-know

- Alternator principle and emf equation
- Synchronous speed
- AC motor classification
- Single-phase and three-phase induction motor working

## Important definitions

| Term                | Quick meaning                                                |
|---------------------|--------------------------------------------------------------|
| Impedance           | Total opposition offered by an AC circuit.                   |
| Power factor        | Cosine of phase angle between voltage and current.           |
| Resonance           | Condition in RLC circuit when $X_L = X_C$ .                  |
| Thevenin equivalent | Voltage source $V_{th}$ in series with resistance $R_{th}$ . |
| Back emf            | Emf induced in DC motor armature opposing applied voltage.   |
| Synchronous speed   | Speed of rotating magnetic field in an AC machine.           |

## Common exam mistakes

- Writing only formula without explaining each symbol.
- Forgetting units in numerical answers.

- Confusing inductive and capacitive phase relation.
- Writing theorem statement but not writing solving steps.
- Drawing machine diagrams without labels.
- Writing transformer works on DC supply. Transformer requires AC/changing flux.

## Last-minute checklist

- 1 Revise formula bank and write each formula once.
- 2 Practice one impedance problem and one Thevenin problem.
- 3 Redraw transformer, DC generator and three-phase induction motor diagrams.
- 4 Revise applications of all motors.
- 5 Read the Master Answer Key for exact answer point format.

## Master Answer Key

Answers below match the Expected Questions shown in Module 1 to Module 4. The question numbering is kept exactly same.

## Module 1 Answers

### M1-Q1.

**Cycle:** one complete positive and negative variation. **Time period:** time for one cycle.

**Frequency:** cycles per second. **Amplitude:** maximum value. **Phase:** angular position of waveform with respect to reference.

### M1-Q2.

Sine wave changes smoothly and is common in AC generation. Square wave switches between two levels. Triangular wave rises and falls linearly. Sawtooth wave rises gradually and falls suddenly or vice versa. Mention shape, harmonic content and common use.

### M1-Q3.

In pure resistance,  $V$  and  $I$  are in phase and power is consumed. In pure inductance, current lags voltage by  $90^\circ$  and energy is stored in magnetic field. In pure capacitance, current leads voltage by  $90^\circ$  and energy is stored in electric field.

### M1-Q4.

Draw current as reference. Draw  $V_R$  in phase with current,  $V_L$  leading current by  $90^\circ$ , and  $V_C$  lagging current by  $90^\circ$ . Resultant supply voltage is vector sum of  $V_R$  and net reactive voltage.

### M1-Q5.

Active power  $P = VI\cos\phi$  in watts. Reactive power  $Q = VI\sin\phi$  in VAR. Apparent power  $S = VI$  in VA. Power factor is  $\cos\phi$ , the cosine of the angle between voltage and current.

### M1-Q6.

**Given:**  $R = 40\ \Omega$ ,  $X_L = 80\ \Omega$ ,  $X_C = 50\ \Omega$ . Net reactance  $= 30\ \Omega$ .  $Z = \sqrt{(40^2 + 30^2)} = \sqrt{2500} = 50\ \Omega$ . Power factor  $= R/Z = 40/50 = 0.8$ . Since  $X_L > X_C$ , power factor is lagging.

### M1-Q7.

Resonance is the condition in an RLC circuit when  $X_L = X_C$ . At resonance, impedance is minimum in a series circuit and current is maximum. Q factor indicates sharpness of resonance and voltage/current magnification.

### **M1-Q8.**

Steps: identify branches, find branch impedance/admittance, calculate branch currents, resolve currents into active and reactive components, add components, find total current, power factor and power.

## **Module 2 Answers**

### **M2-Q1.**

Ohm's law: current through a conductor is directly proportional to voltage when physical conditions are constant,  $V = IR$ . KCL: algebraic sum of currents at a node is zero. KVL: algebraic sum of voltages around a closed loop is zero.

### **M2-Q2.**

Mesh analysis uses loop currents and KVL. It is suitable for planar circuits. Node analysis uses node voltages and KCL. It is useful when current sources are present and the circuit has fewer essential nodes.

### **M2-Q3.**

Superposition theorem states that in a linear bilateral network with multiple independent sources, current or voltage in an element is the algebraic sum of effects produced by each source acting alone. While considering one source, replace other voltage sources by short circuit and current sources by open circuit.

### **M2-Q4.**

Thevenin's theorem: any linear two-terminal network can be replaced by an equivalent voltage source  $V_{th}$  in series with resistance  $R_{th}$ . Steps: remove load, find open-circuit voltage  $V_{th}$ , deactivate independent sources and find  $R_{th}$ , reconnect load to equivalent circuit and solve.

### M2-Q5.

Maximum Power Transfer theorem states that maximum power is delivered to load when load resistance equals internal/Thevenin resistance of source network. Condition:  $R_L = R_{th}$ .

### M2-Q6.

A transformer works on mutual induction. AC in primary creates alternating flux in core. This flux links secondary winding and induces emf. Construction includes laminated core, primary winding, secondary winding, insulation and cooling arrangement.

### M2-Q7.

For sinusoidal flux, RMS induced emf per winding is  $E = 4.44 f N \Phi_m$ . For primary,  $E_1 = 4.44 f N_1 \Phi_m$ . For secondary,  $E_2 = 4.44 f N_2 \Phi_m$ . Explain  $f$  = frequency,  $N$  = turns,  $\Phi_m$  = maximum flux.

### M2-Q8.

Losses: copper loss, hysteresis loss, eddy current loss and leakage losses. Types: step-up, step-down, isolation, core type, shell type, power and distribution transformers. Applications: power transmission, adapters, isolation, measurement and electronic power supplies.

## Module 3 Answers



### M3-Q1.

A DC generator works on Faraday's law. When armature conductors rotate in a magnetic field, they cut magnetic flux and emf is induced. The commutator converts internally alternating emf into unidirectional DC at the brushes.

### M3-Q2.

Generated emf  $E = \frac{P\Phi ZN}{60A}$ , where  $P$  = number of poles,  $\Phi$  = flux per pole,  $Z$  = total armature conductors,  $N$  = speed in rpm,  $A$  = number of parallel paths.

### M3-Q3.

Armature reaction is the effect of armature flux on the main field flux. It distorts and weakens main flux, shifts neutral plane and can cause sparking at brushes. It is reduced using interpoles or compensating windings.

### M3-Q4.

A DC motor works on the principle that a current-carrying conductor placed in a magnetic field experiences force. The direction of force is given by Fleming's left-hand rule. This force produces torque and rotates the armature.

### M3-Q5.

Back emf is the emf induced in the rotating armature of a DC motor that opposes the applied voltage. Significance: it limits armature current during running and makes the motor self-regulating with load.

### M3-Q6.

At starting, back emf is zero, so armature current can become very high because armature resistance is small. A starter adds external resistance at starting to limit current and provides protection.

### M3-Q7.

A 3-point starter has line, armature and field terminals. Starting resistance is gradually cut out as motor speed and back emf increase. It includes no-voltage release and overload release for protection.

### M3-Q8.

DC shunt motor: nearly constant speed, used in fans and lathes. DC series motor: high starting torque, used in traction and cranes. DC compound motor: combines good torque and speed regulation, used in elevators and heavy machines.

## Module 4 Answers

### M4-Q1.

An alternator converts mechanical energy into AC electrical energy. The rotor field rotates and cuts stator conductors, inducing alternating emf in stator windings according to electromagnetic induction.

### M4-Q2.

Alternator emf per phase  $E = 4.44 k_w f \Phi T$ , where  $k_w$  = winding factor,  $f$  = frequency,  $\Phi$  = flux per pole and  $T$  = series turns per phase.

### M4-Q3.

Synchronous speed is the speed of rotating magnetic field. In one mechanical revolution,  $P/2$  cycles are produced. Hence  $f = PN/120$  and  $N_s = 120f/P$  rpm.

### M4-Q4.

Open circuit characteristic of an alternator is the graph between generated emf and field current at constant speed with no load connected. It shows magnetization behavior and saturation.

#### **M4-Q5.**

AC motors can be classified as synchronous motors and induction motors. Induction motors may be single-phase or three-phase. Special AC motors include stepper motor, servo motor and universal motor.

#### **M4-Q6.**

Stepper motor moves in fixed angular steps and is used in positioning. Servo motor uses feedback for precise control and is used in automation. Universal motor works on AC or DC, has high speed and is used in portable tools and appliances.

#### **M4-Q7.**

A single-phase induction motor works by induction but is not self-starting because single-phase supply produces pulsating magnetic field. Starting methods such as split-phase or capacitor start create phase difference to start rotation.

#### **M4-Q8.**

A three-phase induction motor works on rotating magnetic field. Three-phase stator supply creates rotating field, which induces current in rotor conductors. Interaction of rotor current and rotating field produces torque. Applications include pumps, compressors, conveyors and industrial drives.